

MDT-UTN-CBT

Metric-Differential-Tree UTN and Context-Bound Token Intelligence

v1.0.0 | Algorithmic Foundation Release | DOI Repository Overview

Core Thesis

Context is primary; naming is secondary.

UTN is reframed as context-preserving structural localization, folding, typing, and controlled naming over a Metric-Differential Tree.

Canonical Structural Path



Six Core Contributions

1. MDT-UTN

Context -> GenericContainerStarmap -> metric localization
-> MDT -> Per-Node UTN.
2. Typing First

Type before name; preserve leaf identity, provenance, counter-evidence, and explicit leftovers.
3. Composable UTN

Fold locally, merge folded trees globally, and unfold only where ambiguity requires it.
4. Context-Bound Token

Expose structural identity as token@context while preserving the raw-token baseline.
5. Baseline-Preserved Injection

Every context-enhanced input keeps a recoverable context-free control for measuring Context Gain.
6. Context as Address

Use validated context identity to address search, cache, Delta Intelligence, CCC, and Brain Units.

Validation Methodology

VALIDATION-001: Does explicit context help? - Baseline-Preserved Context Injection
VALIDATION-002: Which context helps? - Domain / Space / Time context matrix
VALIDATION-003: How much context helps? - Context Density Policy protocol

Repository Contents

6 core research articles | 3 validation protocols | 7 canonical figures
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Why It Matters

The framework connects structural typing, contextual encoding, prompt/query localization, caching, experience reuse, specialist dispatch, and continual structural growth through one common object: context identity.